

ABG Interpretation Workshop

Faculty Detailed Guide

How this session runs

All 13 cases handed out. 8-minute settling period. Each case = 15-minute unit.

Small Group - 8 min - Group faculty	Plenary - 7 min - Moderator
0-3 min: Team interprets independently 3-6 min: Faculty-mediated discussion 6-8 min: Group synthesises answer	0-3 min: Random draw - one group presents 3-6 min: Case-specific Q and A plus post-intervention gas 6-7 min: Take-home pearl

Timing discipline: hold the 3-3-2 boundary. Resist answering in the first 3 minutes. If the group finishes early, ask a what-if variant.

Case 1 twist: small-group phase works presenting gas only; moderator reveals the 14-hour repeat gas in plenary.

Quick Answer Index - all 13 cases

#	Diagnosis	Moderator take-home
1	DKA	Recalculate the anion gap on every repeat gas - a persistently low HCO_3 after treatment can mean the acid has changed character.
2	Pyloric stenosis	A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.
3	Gastroenteritis	A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.
4	Suspected IEM	Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.
5	Salicylate poisoning	A near-normal pH never means a normal gas - check PaCO_2 and HCO_3 against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.
6	Post-arrest hanging	When PaCO_2 and HCO_3 are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.
7	Septic shock + severe anaemia	A normal SpO_2 says nothing about oxygen delivery when haemoglobin is this low - look at ScvO_2 and lactate, not the saturation monitor. SID reveals the chloride-driven component.
8	Methaemoglobinaemia	A high PaO_2 does not guarantee oxygen-carrying capacity - whenever SpO_2 and PaO_2 disagree, request co-oximetry before you do anything else.
9	Two encephalopathic neonates	Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.
10	HUS-like	HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.
11	Hepatitis A encephalopathy	Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.
12	Fasting hypoglycaemia	Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.
13	Shocked 2-week-old boy	Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.

The Unified 10-Step Approach

Step	Tool / key question
1. Clinical history	What does the story alone suggest?
2. Internal validity	$[H^+] = 24 \times PaCO_2 / HCO_3^-$ - does it match pH? Arterial?
3. pH	Acidaemia, alkalaemia, or normal? A normal pH does not mean a normal gas.
4. Primary disorder	Does $PaCO_2$ or HCO_3^- explain the pH direction?
5. Expected compensation	Metabolic acidosis: $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$. Metabolic alkalosis: $PaCO_2 = 0.7 \times HCO_3^- + 21 \pm 2$.
6. Anion gap	$Na^+ - (Cl^- + HCO_3^-)$; add 2.5 per 1 g/dL albumin below 4.0.
7. Delta ratio	$(AG - 12) / (24 - HCO_3^-)$ - only if AG raised.
8. Oxygenation & integration	P/F, OI, CaO_2 , $ScvO_2$, MetHb. Does the story fit the child?
9. SID / Stewart analysis	SID = $Na^+ - Cl^-$ (normal ~36). Low = chloride-driven acidosis. High = chloride-loss alkalosis. Atot = albumin + phosphate. Unmeasured anions = $AG - (lactate + ketones)$.
10. Post-intervention re-evaluation	Has the mechanism changed? Recalculate AG, SID, Atot.

Case 1 - DKA

Small-group phase (8 min) - you run this

Vignette. 10-year-old girl, 2 weeks of polyuria and 3 days of vomiting and abdominal pain. Acidotic breathing present, haemodynamically stable, GCS 11.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.08	14	4	36	132
K	Cl	Albumin	Glucose	b-OHB
4.5	98	4.0	480	5.8

Worked interpretation

Step	Answer
1. Clinical history	2 weeks of polyuria (osmotic diuresis) with vomiting and abdominal pain, Kussmaul breathing, GCS 11 - classical new-onset DKA with early obtundation.
2. Internal validity	$[H^+] = 24 \times 14 / 4 = 84 \text{ nmol/L} \rightarrow \text{pH } 7.08$ - matches. PaO ₂ 36 suggests a venous sample.
3. pH	7.08 - severe acidaemia.
4. Primary disorder	HCO ₃ 4 (low) with acidaemia $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $\text{PaCO}_2 = 1.5 \times 4 + 8 = 14 \pm 2$. Actual 14 - appropriate.
6. Anion gap	$\text{AG} = 132 - (98 + 4) = 30$ (markedly raised). Albumin normal - no correction needed.
7. Delta ratio	$(30 - 12) / (24 - 4) = 0.9 \rightarrow$ pure HAGMA.
8. Oxygenation & integration	Severe HAGMA with maximal appropriate respiratory compensation; AG 30 fully explained by ketoacids (b-OHB 5.8). Severe DKA.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal - mild chloride-driven component already present.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$30 - 5.8 = 24.2$	Markedly raised - ketoacids dominate.
Mechanism	-	Unmeasured anions, with mild chloride-driven component.

Where groups go wrong

- Calling this respiratory alkalosis from the low PaCO₂ alone.
- Missing that the low PaO₂ (venous sample) is itself a validity teaching point.
- Forgetting to correct the anion gap for albumin as a reflex step.

Your synthesis (final 2 min)

Severe DKA, appropriately compensated, pure HAGMA. Plan: fluid resuscitation, insulin infusion with potassium replacement anticipated.

Plenary phase (7 min) - moderator runs this

Case-specific questions

Q1. Calculate the corrected serum sodium.

Q2. What do the glucose and b-OHB values suggest?

Answers

A1. $\text{Na}^+ + 1.6 \times [(480 - 100)/100] \sim 138$. True hyponatraemia is dilutional from hyperglycaemia.

A2. Hyperglycaemic ketosis driving the gap - severe DKA.

Post-intervention reveal

Intervention. 10 mL/kg NS bolus, insulin infusion, NS + KCl.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	34	142
K	Cl	Albumin	Glucose	b-OHB
3.4	119	4.0	230	0.6

Step	Answer
2. Internal validity	[H ⁺] ~56 nmol/L - internally valid.
3. pH	7.25 - improving.
4. Primary disorder	HCO ₃ 11 -> metabolic acidosis.
5. Expected compensation	Winter's: PaCO ₂ = 1.5 x 11 + 8 = 24.5 +/-2. Actual 26 - appropriate.
6. Anion gap	AG = 142 - (119 + 11) = 12 (normal - HAGMA resolved).
7. Delta ratio	(12 - 12)/(24 - 11) = 0 -> pure NAGMA.
9. SID / Stewart	SID = 142 - 119 = 23 (markedly low - chloride-driven). Unmeasured anions = 12 - 0.6 = 11.4 (normal). Mechanism changed: ketoacids cleared, chloride load now driving.

Post-intervention case-specific answers

Q1. Compare this ABG with the initial ABG. What has changed?

Q2. Why is acidosis persisting despite improvement in glucose and b-OHB?

Q3. Calculate the corrected serum sodium.

A1. pH improved (7.08->7.25); PaCO₂ rose (14->26); HCO₃ rose modestly (4->11); glucose fell (480->230); b-OHB fell (5.8->0.6); Na rose (132->142); K fell (4.5->3.4); Cl rose sharply (98->119); AG normalised (30->12). The acid has changed character.

A2. Ketoacidosis resolved. Persisting low HCO₃ is now hyperchloraemic NAGMA from saline chloride load - expected, not treatment failure.

A3. $142 + 1.6 \times [(230 - 100)/100] \sim 144$.

Moderator take-home (final 1 min)

Recalculate the anion gap on every repeat gas - a persistently low HCO₃ after treatment can mean the acid has changed character.

Case 2 - Pyloric stenosis

Small-group phase (8 min) - you run this

Vignette. 6-week-old first-born boy with 10 days of progressively forceful, non-bilious vomiting after every feed. Sunken eyes, lethargic but arousable, still hungry after vomiting.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.52	48	38	34	132
K	Cl	Albumin	Glucose	Urine Cl
3.0	82	4.0	90	<15

Worked interpretation

Step	Answer
1. Clinical history	Classic for hypertrophic pyloric stenosis: forceful non-bilious vomiting after every feed, still hungry, with dehydration.
2. Internal validity	$[H^+] = 24 \times 48 / 38 = 30 \rightarrow$ pH 7.52 - matches. PaO ₂ 34 suggests venous.
3. pH	7.52 - alkalaemia.
4. Primary disorder	HCO ₃ 38 (raised) with alkalaemia $\rightarrow$ metabolic alkalosis.
5. Expected compensation	$PaCO_2 = 0.7 \times 38 + 21 = 47.6 \pm 2$ (45.6-49.6). Actual 48 - appropriate.
6. Anion gap	$AG = 132 - (82 + 38) = 12$ (normal).
7. Delta ratio	Not applicable - no HAGMA.
8. Oxygenation & integration	Hypochloraemic, hypokalaemic metabolic alkalosis with appropriate respiratory compensation; low urine Cl confirms chloride-responsive process.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 82 = 50$	Markedly high - chloride-loss alkalosis.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$AG\ 12 - 0 = 12$	Normal.
Mechanism	-	Loss of gastric HCl has removed chloride, raising SID.

Where groups go wrong

- Trying to apply the delta ratio or AG framework to an alkalosis.
- Overlooking K 3.0 as an urgent, actionable finding.

Your synthesis (final 2 min)

A textbook gas: use it to drill the metabolic-alkalosis compensation formula and the urine-chloride distinction.

Plenary phase (7 min) - moderator runs this

Case-specific questions

Q1. What is the significance of the low urine chloride?

Q2. What is the likely clinical diagnosis?

Answers

A1. Urine Cl < 15-20 mmol/L confirms chloride-responsive metabolic alkalosis: kidney appropriately conserving chloride. Alkalosis will correct with chloride-rich fluid. Contrast with chloride-resistant causes (Bartter/Gitelman) where urine Cl is inappropriately high.

A2. Hypertrophic pyloric stenosis - persistent vomiting of gastric HCl and volume depletion produce hypochloraemic, hypokalaemic metabolic alkalosis; secondary hyperaldosteronism drives further renal H⁺ and K⁺ loss.

Post-intervention reveal

Intervention. IV 0.9% saline + KCl, correct dehydration and hypokalaemia, then pyloromyotomy.

pH	PaCO ₂	HCO ₃	Na	K
7.45	40	27	138	4.0
Cl	Albumin	Urine Cl		
100	4.0	>20		

Step	Answer
9. SID / Stewart	SID = 138 - 100 = 38 (normalising - chloride repletion working). Urine Cl now >20 confirms kidney no longer conserving chloride.

Moderator take-home (final 1 min)

A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.

Case 3 - Gastroenteritis

Small-group phase (8 min) - you run this

Vignette. 8-month-old girl with 2 days of fever and large-volume loose stools. Sunken eyes, delayed skin turgor, prolonged capillary refill time.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.22	27	11	34	129
K	Cl	Albumin	Glucose	Lactate
3.2	112	3.5	85	1

Worked interpretation

Step	Answer
1. Clinical history	Acute gastroenteritis with large-volume diarrhoea and significant dehydration.
2. Internal validity	$[H^+] = 24 \times 27 / 11 = 59 \rightarrow$ pH 7.23 - matches. PaO ₂ 34 suggests venous.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$ (22.5-26.5). Actual 27 is at upper edge - borderline compensation.
6. Anion gap	$AG = 129 - (112 + 11) = 6$ (low/normal). Corrected $AG = 6 + 2.5 \times (4 - 3.5) = 7.25$ - still normal.
7. Delta ratio	Not applicable - no raised AG.
8. Oxygenation & integration	NAGMA (hyperchloraemic) from stool bicarbonate loss; normal lactate argues against hypoperfusion.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$129 - 112 = 17$	Markedly low - severe chloride-driven acidosis.
Atot	Albumin 3.5	Mildly low - slightly alkalinising.
Unmeasured anions	$AG\ 6 - 1 = 5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Where groups go wrong

- Expecting a raised anion gap because the child is dehydrated.
- Not pairing with Case 4 as a contrast: correction reveals nothing hidden here.

Your synthesis (final 2 min)

Pair explicitly with Case 4 - both have normal-looking raw AG and mild hypoalbuminaemia, but correction reveals nothing hidden here, whereas in Case 4 it unmasks a concealed HAGMA.

Plenary phase (7 min) - moderator runs this

Case-specific questions

Q1. What is the likely cause of the metabolic acidosis in this child?

Q2. How do the chloride and lactate values help you identify the mechanism?

Answers

A1. Loss of bicarbonate-rich fluid in large-volume diarrhoea $\rightarrow$ hyperchloraemic NAGMA.

A2. Chloride elevated (112) as it replaces lost bicarbonate; normal lactate (1) argues against significant hypoperfusion/shock.

Post-intervention reveal

Intervention. IV isotonic fluids, correct dehydration.

pH	PaCO ₂	HCO ₃	Na	K
7.35	36	20	135	4.0
Cl	Albumin	Lactate		
105	3.5	1		

Step	Answer
6. Anion gap	AG = 135 - (105 + 20) = 10 - normal.
9. SID / Stewart	SID = 135 - 105 = 30 (improving).

Moderator take-home (final 1 min)

A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.

Case 4 - Suspected IEM

Small-group phase (8 min) - you run this

Vignette. 8-month-old girl with 1 day of fever and vomiting. Effortless tachypnoea, haemodynamically stable, lethargic. History of similar episodes in the past.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.24	38	16	38	135
K	Cl	Albumin	Glucose	Lactate
4	107	2	85	3

Worked interpretation

Step	Answer
1. Clinical history	Short acute illness in an infant with history of similar prior episodes - recurrence is the key word.
2. Internal validity	$[H+] = 24 \times 38 / 16 = 57 \rightarrow$ pH 7.24 - matches.
3. pH	7.24 - acidaemia.
4. Primary disorder	HCO ₃ 16 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 16 + 8 = 32 \pm 2$ (30-34). Actual 38 - higher than expected $\rightarrow$ additional respiratory acidosis.
6. Anion gap	Raw AG = $135 - (107 + 16) = 12$ (apparently normal). Albumin 2.0: corrected AG = $12 + 2.5 \times (4 - 2) = 17$ (clearly raised).
7. Delta ratio	Using corrected AG: $(17 - 12) / (24 - 16) = 5/8 = 0.625 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	Concealed HAGMA unmasked only after albumin correction, with inadequate respiratory compensation, in child with recurrent episodes $\rightarrow$ suspect IEM.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$135 - 107 = 28$	Low - chloride-driven component.
Atot	Albumin 2.0	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17 - 3 = 14$	Raised.
Mechanism	-	Three forces: low SID, low Atot (masking), unmeasured anions.

Where groups go wrong

- Stopping at AG 12, normal, without applying the albumin correction - the single most consequential miss in the whole case set.
- Treating lactate 3 as fully explaining the anion-gap elevation.

Your synthesis (final 2 min)

Correct the anion gap for albumin as a reflex, every time, especially when the raw number looks reassuring. Here it changes the entire diagnostic direction.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the albumin-corrected anion gap.
- Q2. Is there an additional respiratory disorder?
- Q3. What is the significance of recurrent similar episodes in this clinical setting?

Answers

A1. $12 + 2.5 \times (4.0 - 2.0) = 17$ mEq/L. Raw gap of 12 looked normal; low albumin was masking a real HAGMA.

A2. Yes - Winter's predicts PaCO₂ 30-34; actual 38 is higher -> inadequate compensation (superimposed respiratory acidosis).

A3. Recurrent metabolic decompensation triggered by intercurrent illness raises strong suspicion for underlying IEM (organic acidaemia or fatty acid oxidation defect). Prompt metabolic work-up: ammonia, lactate, urine organic acids, plasma amino acids, acylcarnitine - the critical sample.

Post-intervention reveal

Intervention. Critical sample, dextrose, stop protein.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	20	138	4.2
Cl	Albumin	Glucose	Lactate	
105	2.5	100	2	

Step	Answer
9. SID / Stewart	SID = $138 - 105 = 33$ (normalising). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG ($20 + 2.5 \times 1.5 = 23.75$) - 2 = 21.75 (still raised).

Moderator take-home (final 1 min)

Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.

Case 5 - Salicylate poisoning

Small-group phase (8 min) - you run this

Vignette. 15-year-old boy with tinnitus, fast breathing, vomiting, and confusion after intentional ingestion.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.42	18	11	108	140
K	Cl	Albumin	Glucose	Lactate
4.2	102	4.1	90	3.2

Worked interpretation

Step	Answer
1. Clinical history	Tinnitus, hyperventilation, vomiting, confusion after intentional ingestion in a teenager - tinnitus is the most specific clue.
2. Internal validity	$[H^+] = 24 \times 18 / 11 = 39 \rightarrow$ pH 7.41 - matches 7.42. Plausible arterial sample.
3. pH	7.42 - essentially normal (the trap).
4. Primary disorder	Two markedly abnormal primary values (PaCO ₂ 18, HCO ₃ 11) with near-normal pH - mixed disorder.
5. Expected compensation	If HCO ₃ 11 were the only problem, Winter's predicts PaCO ₂ 24.5 +/-2. Actual PaCO ₂ 18 is far below that - separate primary respiratory alkalosis.
6. Anion gap	AG = 140 - (102 + 11) = 27 (markedly raised). Lactate only partly explains.
7. Delta ratio	$(27 - 12)/(24 - 11) = 1.15 \rightarrow$ HAGMA.
8. Oxygenation & integration	Primary respiratory alkalosis + primary HAGMA, opposing pH effects cancelling to near-normal pH - classic salicylate poisoning.

SID / Stewart analysis

Tool	Value	Interpretation
SID	140 - 102 = 38	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	27 - 3.2 = 23.8	Markedly raised - salicylate metabolites + ketones.
Mechanism	-	Unmeasured anions + primary respiratory alkalosis.

Where groups go wrong

- Seeing pH 7.42 and concluding the gas is essentially normal.
- Attributing low PaCO₂ purely to appropriate compensation without checking against Winter's.

Your synthesis (final 2 min)

This case exists to break the habit of reading the pH first and stopping. Always check PaCO₂ and HCO₃ against each other.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1.** A near-normal pH can conceal a mixed disorder. What disorders are present here?
- Q2.** What poisoning is suggested by the clinical history and ABG pattern?

Answers

A1. Primary respiratory alkalosis (PaCO₂ far lower than metabolic compensation alone would predict - direct salicylate stimulation of central respiratory centre) + primary HAGMA (AG 27, from lactate, ketosis, salicylate organic acids).

A2. Salicylate (aspirin) poisoning - classic triad of tinnitus, hyperventilation, and mixed respiratory-alkalosis/HAGMA pattern after intentional ingestion in an adolescent.

Post-intervention reveal

Intervention. Activated charcoal, IV fluids, sodium bicarbonate, alkalinise urine, consider dialysis.

pH	PaCO2	HCO3	Na	K
7.48	30	22	142	3.8
Cl	Albumin	Lactate		
105	4.1	1.5		

Step	Answer
9. SID / Stewart	SID = 142 - 105 = 37 (normal). Unmeasured anions = AG (142-105-22 = 15) - 1.5 = 13.5 (improving).

Moderator take-home (final 1 min)

A near-normal pH never means a normal gas - check PaCO2 and HCO3 against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 6 - Post-arrest hanging

Small-group phase (8 min) - you run this

Vignette. 5-year-old boy, 20 kg, accidental hanging. Post-arrest, ROSC after 15 min CPR, ventilated. PEEP 6, PIP 16, RR 20, I:E 1:2, FiO2 100%, TV 80 mL, MAP 17.

Presenting gas

pH	PaCO2	HCO3	PaO2	Na
6.88	65	12	70	138
K	Cl	Albumin	Glucose	Lactate
6.2	98	4.1	210	3.2

Worked interpretation

Step	Answer
1. Clinical history	Post-cardiac-arrest after prolonged CPR for hanging - expect hypoxic-ischaemic metabolic acidosis and possible inadequate ventilation.
2. Internal validity	$[H^+] = 24 \times 65 / 12 = 130 \rightarrow \text{pH } 6.89$ - matches 6.88.
3. pH	6.88 - profound, life-threatening acidaemia.
4. Primary disorder	Both HCO3 12 and PaCO2 65 abnormal in acidifying direction - combined metabolic + respiratory acidosis.
5. Expected compensation	If pure metabolic, Winter's predicts PaCO2 26 +/-2. Actual 65 far above - large separate primary respiratory acidosis.
6. Anion gap	$AG = 138 - (98 + 12) = 28$ (markedly raised). Lactate 3.2 does not fully explain.
7. Delta ratio	$(28 - 12) / (24 - 12) = 1.33 \rightarrow$ HAGMA component.
8. Oxygenation & integration	$P/F = 70/1.0 = 70$. $OI = (17 \times 1.0 \times 100)/70 = 24.3$. Severe ARDS (PALICC OI ≥ 16).

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 98 = 40$	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$28 - 3.2 = 24.8$	Markedly raised - hypoxic-ischaemic anions.
Mechanism	-	Unmeasured anions + primary respiratory acidosis.

Where groups go wrong

- Assuming high PaCO2 is simply the ventilator not yet caught up without naming it as a second primary disorder.
- Not flagging K 6.2 as immediate actionable safety issue.
- Not noticing lactate 3.2 under-explains AG 28.

Your synthesis (final 2 min)

Most acidotic gas in the set exists to force explicit separation of the two acidoses and bring in oxygenation indices.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Identify all components of the mixed acid-base disorder.
- Q2. Calculate the P/F ratio and oxygenation index using the available data.
- Q3. How would you grade the severity of oxygenation failure?

Answers

A1. (1) Primary metabolic acidosis (HCO3 12, AG 28 - hypoxic-ischaemic/lactic) and (2) primary respiratory acidosis (PaCO2 65, far exceeding expected ~26) - combined metabolic + respiratory acidosis.

A2. $P/F = 70/1.0 = 70$. $OI = (17 \times 1.0 \times 100)/70 = 24.3$.

A3. $OI\ 24.3$ = severe oxygenation failure (PALICC $OI \geq 16$ = severe).

Post-intervention reveal

Intervention. Increase ventilation, treat hyperkalaemia, fluids, consider bicarbonate.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	40	17	90	138
K	Cl	Albumin	Glucose	Lactate
4.5	100	4.1	180	2.5

Step	Answer
9. SID / Stewart	$SID = 138 - 100 = 38$ (normal). Unmeasured anions = $AG\ (138 - 100 - 17 = 21) - 2.5 = 18.5$ (still raised).

Moderator take-home (final 1 min)

When PaCO₂ and HCO₃ are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.

Case 7 - Septic shock + severe anaemia

Small-group phase (8 min) - you run this

Vignette. 4-year-old boy with septic shock and severe anaemia. FiO₂ 40%, SpO₂ 99%, ScvO₂ 42%.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	110	138
K	Cl	Albumin	Hb	Lactate
4.3	106	4.1	4.5	7

Worked interpretation

Step	Answer
1. Clinical history	Septic shock with severe anaemia (Hb 4.5) - two independent reasons for impaired oxygen delivery.
2. Internal validity	$[H^+] = 24 \times 26 / 11 = 57 \rightarrow$ pH 7.25 - matches.
3. pH	7.25 - acidaemia.
4. Primary disorder	HCO ₃ 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$. Actual 26 - appropriate.
6. Anion gap	$AG = 138 - (106 + 11) = 21$ (raised). Albumin normal, negligible correction.
7. Delta ratio	$(21 - 12) / (24 - 11) = 0.69 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	$P/F = 110 / 0.4 = 275$. ScvO ₂ 42% - tissues extracting far more O ₂ than usual. $CaO_2 = (1.34 \times 4.5 \times 0.99) + (0.003 \times 110) = 6.3$ mL/dL.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 106 = 32$	Low - chloride-driven component from resuscitation fluids.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$21 - 7 = 14$	Raised - lactic acidosis.
Mechanism	-	Two forces: low SID + unmeasured anions.

Where groups go wrong

- Reassurance from normal SpO₂/PaO₂ without looking at ScvO₂.
- Forgetting that severe anaemia is, by itself, full explanation for impaired oxygen delivery.

Your synthesis (final 2 min)

Anchor on oxygen-delivery equation ($DO_2 = \text{cardiac output} \times CaO_2$, and CaO_2 driven mainly by $Hb \times SO_2$).

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the P/F ratio.
- Q2. How do you interpret the low ScvO₂ despite a normal SpO₂?
- Q3. What do the severe anaemia and elevated lactate indicate about oxygen delivery and extraction?

Answers

- A1. $110 / 0.4 = 275$ - mild oxygenation impairment; not where the problem lies.
- A2. Large veno-arterial oxygen extraction gap: lungs saturating haemoglobin normally, but tissues extracting abnormally high fraction because overall DO_2 inadequate relative to demand. SpO₂ alone misses this.

A3. Severe anaemia critically reduces CaO₂ and therefore DO₂, even though saturation normal. Body compensates initially by increasing extraction (low ScvO₂), but once exceeded, anaerobic metabolism follows - lactate ↑. Content/delivery problem, not lung problem. Priority: transfusion.

Post-intervention reveal

Intervention. Blood transfusion, antibiotics, fluids, vasopressors.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.35	36	20	100	138
K	Cl	Albumin	Hb	Lactate
4.0	105	3.8	9.0	2.5
ScvO ₂				
65%				

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-20 = 13) - 2.5 = 10.5 (normalising).

Moderator take-home (final 1 min)

A normal SpO₂ says nothing about oxygen delivery when haemoglobin is this low - look at ScvO₂ and lactate, not the saturation monitor. SID reveals the chloride-driven component.

Case 8 - Methaemoglobinaemia

Small-group phase (8 min) - you run this

Vignette. 10-year-old boy with cyanosis, headache, tachycardia, and lethargy after receiving topical benzocaine. Blood sample drawn looks chocolate-brown. FiO2 90%, SpO2 85%, SO2 65%.

Presenting gas

pH	PaCO2	HCO3	PaO2	Na
7.36	38	21	550	138
K	Cl	Albumin	MetHb	Lactate
4.3	105	4.1	35%	2

Worked interpretation

Step	Answer
1. Clinical history	Cyanosis, headache, tachycardia, lethargy after topical benzocaine, with chocolate-brown blood - close to diagnostic before numbers read.
2. Internal validity	$[H^+] = 24 \times 38 / 21 = 43 \rightarrow$ pH 7.36 - matches. Essentially normal acid-base gas.
3. pH	7.36 - essentially normal.
4. Primary disorder	No significant primary acid-base disorder. Acid-base is distraction.
5. Expected compensation	Not applicable.
6. Anion gap	$AG = 138 - (105 + 21) = 12$ (normal).
7. Delta ratio	Not applicable.
8. Oxygenation & integration	$P/F = 550/0.9 = 611$ (remarkably high). Yet SpO2 85%, SO2 65% - large discordant saturation gap. MetHb 35% (normal <1-2%).

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$12 - 2 = 10$	Normal. Acid-base status benign.
Mechanism	-	Acid-base is not the story; oxygenation is.

Where groups go wrong

- Getting drawn into acid-base analysis when the gas is essentially normal.
- Assuming a high PaO2 guarantees adequate oxygen delivery.

Your synthesis (final 2 min)

Close with the rule this case exists to teach: a normal or high PaO2 does not guarantee normal oxygen-carrying capacity.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the P/F ratio.
- Q2. How do you explain the discrepancy between the very high PaO2, SpO2, and SO2?
- Q3. What is the significance of the chocolate-brown blood and MetHb 35%?
- Q4. What is the most likely diagnosis?

Answers

A1. $550 / 0.9 \sim 611$.

A2. PaO2 measures dissolved oxygen tension and is unaffected by methaemoglobin, so remains high. Methaemoglobin (Fe3+, unable to bind oxygen) distorts pulse oximeter algorithm, driving SpO2 toward falsely fixed value ~82-85%. Directly measured SO2 by co-oximetry (65%) reflects true, much lower functional saturation - 35% of haemoglobin cannot carry oxygen at all.

A3. Chocolate-brown blood is classic bedside sign. MetHb 35% confirms severe elevation - enough to cause significant functional anaemia (only 65% of haemoglobin can carry oxygen). Explains cyanosis, headache, tachycardia, lethargy despite high PaO2.

A4. Acquired methaemoglobinaemia (benzocaine-induced).

Post-intervention reveal

Intervention. Methylene blue 1-2 mg/kg IV (check G6PD status first), oxygen, stop benzocaine.

pH	PaCO2	HCO3	PaO2	Na
7.38	38	23	200	138
K	Cl	Albumin	MetHb	Lactate
4.0	105	4.1	5%	1.5
SpO2	SO2			
98%	95%			

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-23 = 10) - 1.5 = 8.5 (normal).

Moderator take-home (final 1 min)

A high PaO2 does not guarantee oxygen-carrying capacity - whenever SpO2 and PaO2 disagree, request co-oximetry before you do anything else.

Case 9 - Two encephalopathic neonates

Small-group phase (8 min) - you run this

Vignette. Baby A: 4 days old, fed well until day 2, now lethargic and tachypnoeic. Afebrile, clear chest. On antibiotics for sepsis. Baby B: 7 days old, vomiting, lethargic, deep rapid breathing. Platelets 60,000/uL, urine ketones 3+.

Presenting gas

A pH	A PaCO ₂	A HCO ₃	A SBE	A Na/Cl
7.51	26	20	-1	138/104
A Glucose	A Lactate	B pH	B PaCO ₂	B HCO ₃
85	1.8	7.14	18	6
B SBE	B Na/Cl	B Glucose	B Lactate	
-21	136/100	45	3.0	

Worked interpretation

Step	Answer
1. Clinical history	A: fed well until day 2, now lethargic and tachypnoeic; sepsis label but afebrile, clear chest. B: vomiting, lethargic, deep rapid breathing, thrombocytopenia, ketonuria.
2. Internal validity	A: $[H^+] = 24 \times 26 / 20 = 31 \rightarrow$ pH 7.51 - matches. B: $[H^+] = 24 \times 18 / 6 = 72 \rightarrow$ pH 7.14 - matches.
3. pH	A: 7.51 - alkalaemia. B: 7.14 - acidaemia.
4. Primary disorder	A: PaCO ₂ 26 (low) $\rightarrow$ respiratory alkalosis. B: HCO ₃ 6 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	A: acute resp alkalosis expected HCO ₃ $\sim$ 21.2; actual 20 - appropriate. B: Winter's predicted PaCO ₂ 17 ± 2 ; actual 18 - appropriate.
6. Anion gap	A: AG = $138 - (104 + 20) = 14$ (normal). B: AG = $136 - (100 + 6) = 30$ (markedly raised).
7. Delta ratio	A: not applicable. B: $(30 - 12)/(24 - 6) = 1.0 \rightarrow$ HAGMA.
8. Oxygenation & integration	A: sepsis alone does not explain respiratory alkalosis - suspect hyperammonaemia. B: lactate only 3.0 - rest of gap filled by ketones and organic acids.

SID / Stewart analysis

Tool	Value	Interpretation
A SID	$138 - 104 = 34$	Low-normal.
A Unmeasured	$14 - 1.8 = 12.2$	Normal. Mechanism: respiratory alkalosis.
B SID	$136 - 100 = 36$	Normal.
B Unmeasured	$30 - 3 = 27$	Markedly raised - organic acids/ketones.

Where groups go wrong

- Attributing Baby A alkalosis to sepsis without considering hyperammonaemia.
- Not checking ammonia on ice for both babies.
- Assuming lactate explains Baby B AG when it only accounts for part.

Your synthesis (final 2 min)

Both babies have a metabolic disorder masquerading as sepsis. The gas is the clue - ammonia on ice is the test.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Name the primary disorder in each baby. Is compensation appropriate in each?
- Q2. Which features does sepsis alone fail to explain?

Q3. Baby B has an anion gap of 30 but lactate is only 3. What fills the rest?

Q4. Intervene: what one test do you send in both, how must the sample be handled, and what do you do while waiting?

Answers

A1. A: respiratory alkalosis, appropriate. B: metabolic acidosis, appropriate.

A2. A: respiratory alkalosis without fever or distress. B: severe HAGMA with ketones and hypoglycaemia.

A3. Ketones (urine 3+) and organic acids from inborn error of metabolism.

A4. Ammonia. Sample on ice, delivered to lab within 15 minutes. While waiting: stop protein, give dextrose, prepare ammonia scavengers.

Post-intervention reveal

Intervention. Both: ammonia level, stop protein, dextrose. A: ammonia scavengers if high. B: carnitine, treat acidosis.

A pH	A PaCO ₂	A HCO ₃	A Na/Cl	A Glucose
7.45	35	22	138/105	90
A Lactate	B pH	B PaCO ₂	B HCO ₃	B Na/Cl
1.5	7.30	25	12	138/102
B Glucose	B Lactate			
100	2.0			

Step	Answer
9. SID / Stewart	A: SID = 138 - 105 = 33; unmeasured anions normal. B: SID = 138 - 102 = 36; unmeasured anions = AG (138-102-12 = 24) - 2 = 22 (still raised).

Moderator take-home (final 1 min)

Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 10 - HUS-like

Small-group phase (8 min) - you run this

Vignette. 3-year-old, 5 days after bloody diarrhoea, now oliguric and pale. Hb 6.2 g/dL, platelets 40,000/uL, creatinine 3.8 mg/dL.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.22	25	10	-16	131
K	Cl	Albumin	Phosphate	iCa
6.8	104	3.0	9.5	0.92

Worked interpretation

Step	Answer
1. Clinical history	Bloody diarrhoea followed by oliguria and pallor, with anaemia, thrombocytopenia, and renal failure - HUS-like picture.
2. Internal validity	$[H+] = 24 \times 25 / 10 = 60 \rightarrow$ pH 7.22 - matches.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 10 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 10 + 8 = 23 \pm 2$. Actual 25 - appropriate.
6. Anion gap	$AG = 131 - (104 + 10) = 17$. Albumin-corrected $AG = 17 + 2.5 \times (4 - 3.0) = 19.5$ (raised).
7. Delta ratio	$(17 - 12)/(24 - 10) = 5/14 = 0.36 \rightarrow$ mixed NAGMA + HAGMA.
8. Oxygenation & integration	Mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$131 - 104 = 27$	Markedly low - chloride-driven.
Atot	Albumin 3.0 (alkalinising), phosphate 9.5 (markedly acidifying)	Net high Atot.
Unmeasured anions	$19.5 - 0 = 19.5$	Raised - urate, oxalate, retained organic anions.
Mechanism	-	Three forces: low SID + high Atot (phosphate) + unmeasured anions.

Where groups go wrong

- Missing that all three Stewart forces are active here.
- Not recognising K 6.8 and iCa 0.92 as immediate actionable threats.
- Reaching for bicarbonate without considering the risks.

Your synthesis (final 2 min)

This is the triple-hit acidosis case. SID, Atot, and unmeasured anions each contribute. Treat the electrolytes first.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1.** Is compensation appropriate? What are the anion gap and the albumin-corrected anion gap?
- Q2.** Calculate the delta ratio. What does it tell you?
- Q3.** Using SID and Atot, name the three separate forces acidifying this child.
- Q4.** Intervene: which two sidekick values change your next 10 minutes? The team asks for bicarbonate - what do you say?

Answers

A1. Appropriate. AG 17, corrected AG 19.5.

A2. 0.36 -> mixed NAGMA + HAGMA.

A3. Low SID (chloride-driven), high Atot (phosphate), unmeasured anions.

A4. K 6.8 (hyperkalaemia) and iCa 0.92 (hypocalcaemia). Bicarbonate? No - pH 7.22 not severe; bicarbonate can worsen hypocalcaemia and hyperkalaemia.

Post-intervention reveal

Intervention. Dialysis, treat hyperkalaemia and hypocalcaemia.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	18	135	4.5
Cl	Albumin	Phosphate	iCa	
105	3.2	6.0	1.0	

Step	Answer
9. SID / Stewart	SID = 135 - 105 = 30 (improving). Atot: phosphate improved from 9.5 to 6.0. Unmeasured anions = AG (135-105-18 = 12) - 0 = 12 (normalising).

Moderator take-home (final 1 min)

HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.

Case 11 - Hepatitis A encephalopathy

Small-group phase (8 min) - you run this

Vignette. 6-year-old with hepatitis A, grade II encephalopathy. INR 4.2, ammonia 180 umol/L.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.49	28	21	-1	132
K	Cl	Albumin	Lactate	Glucose
3.0	98	2.2	5.5	45

Worked interpretation

Step	Answer
1. Clinical history	Hepatitis A with grade II encephalopathy, coagulopathy, hyperammonaemia.
2. Internal validity	$[H+] = 24 \times 28 / 21 = 32 \rightarrow$ pH 7.49 - matches.
3. pH	7.49 - alkalaemia.
4. Primary disorder	PaCO ₂ 28 (low) $\rightarrow$ respiratory alkalosis.
5. Expected compensation	Acute resp alkalosis: HCO ₃ falls ~2 per 10 mmHg fall in PaCO ₂ . PaCO ₂ 28 is 12 below 40 $\rightarrow$ expected HCO ₃ ~21.6. Actual 21 - appropriate.
6. Anion gap	$AG = 132 - (98 + 21) = 13$. Corrected AG = $13 + 2.5 \times (4 - 2.2) = 17.5$ (HAGMA masked by hypoalbuminaemia).
7. Delta ratio	Using corrected AG: $(17.5 - 12) / (24 - 21) = 5.5 / 3 = 1.83 \rightarrow$ mixed.
8. Oxygenation & integration	Respiratory alkalosis + masked HAGMA + hypoglycaemia.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal.
Atot	Albumin 2.2	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17.5 - 5.5 = 12$	Raised.
Mechanism	-	Masked HAGMA + respiratory alkalosis.

Where groups go wrong

- Taking SBE -1 at face value without correcting AG for albumin.
- Missing that the low albumin is masking a HAGMA.
- Not treating hypoglycaemia urgently.

Your synthesis (final 2 min)

The SBE looks normal, but the low albumin is lying to you. Correct the AG, check SID, and treat the glucose.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1.** What is the primary disorder, and why is this child alkaline?
- Q2.** Calculate the anion gap, then correct it for albumin. What changes?
- Q3.** The base excess is -1, which looks normal. Why is that misleading? How many processes are running at once?
- Q4.** Intervene: which two sidekicks need treating in the next few minutes, and why does one of them worsen the encephalopathy?

Answers

- A1.** Respiratory alkalosis due to hyperventilation from ammonia stimulating respiratory centre.

A2. AG 13, corrected AG 17.5. Low albumin was masking a HAGMA.

A3. Yes - because HAGMA is masked by hypoalbuminaemia. Multiple processes: respiratory alkalosis, HAGMA (lactic acidosis), hypoglycaemia.

A4. Glucose 45 (hypoglycaemia) and lactate 5.5 (lactic acidosis). Hypoglycaemia worsens encephalopathy.

Post-intervention reveal

Intervention. Dextrose, lactulose, ammonia scavengers.

pH	PaCO ₂	HCO ₃	Na	K
7.42	32	20	135	3.5
Cl	Albumin	Lactate	Glucose	
100	2.5	3.0	100	

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normal). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG (15 + 3.75 = 18.75) - 3 = 15.75 (still raised).

Moderator take-home (final 1 min)

Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.

Case 12 - Fasting hypoglycaemia

Small-group phase (8 min) - you run this

Vignette. 18-month-old, previously well, fasted overnight after gastroenteritis and presented with a seizure. Liver palpable 3 cm. ALT 180 U/L, ammonia 110 umol/L, free fatty acids raised.

Presenting gas

Glucose	pH	PaCO ₂	HCO ₃	SBE
30	7.36	36	20	-4
Na/Cl 138/105	Lactate 2.0	b-OHB 0.3		

Worked interpretation

Step	Answer
1. Clinical history	Fasting hypoglycaemia with seizure, hepatomegaly, raised ALT and ammonia, raised free fatty acids - key is the inappropriately low ketones.
2. Internal validity	$[H+] = 24 \times 36 / 20 = 43 \rightarrow$ pH 7.36 - matches.
3. pH	7.36 - normal.
4. Primary disorder	HCO ₃ 20 (slightly low), SBE -4 $\rightarrow$ mild metabolic acidosis.
5. Expected compensation	Not applicable (primary metabolic).
6. Anion gap	$AG = 138 - (105 + 20) = 13$ (normal).
7. Delta ratio	Not applicable (NAGMA).
8. Oxygenation & integration	Near-normal gas but hypoglycaemia with low ketones $\rightarrow$ failure of ketogenesis.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin not given	-
Unmeasured anions	$13 - 2 = 11$	Normal.
Mechanism	-	Acidosis mild and chloride-driven; the story is ketogenesis failure.

Where groups go wrong

- Being reassured by the near-normal gas and missing the hypoglycaemia with inappropriately low ketones.
- Drawing the critical sample after giving dextrose.

Your synthesis (final 2 min)

The gas is a distraction. The diagnosis is in the glucose and ketones. Critical sample before dextrose.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. The gas looks almost normal. Why is that the most worrying finding?
- Q2. What should a fasting child gas and ketones look like?
- Q3. Free fatty acids are high but ketones are low. What does that mean? How would hyperinsulinism differ?
- Q4. Intervene: list the critical sample. When exactly must it be drawn, and what do you give immediately afterwards?

Answers

- A1. In a fasting child, expected ketosis and metabolic acidosis. Normal gas with low b-OHB (0.3) indicates failure of ketogenesis.
- A2. Metabolic acidosis with high anion gap due to ketones.

- A3.** Block in fatty acid oxidation or ketogenesis (e.g., MCAD deficiency). Hyperinsulinism would have low FFA and low ketones.
- A4.** Must be drawn during hypoglycaemia before treatment. Includes glucose, insulin, C-peptide, cortisol, growth hormone, ketones, free fatty acids, acylcarnitine profile, ammonia, lactate, LFTs. Give dextrose (2-3 mL/kg of 10% dextrose) immediately after.

Post-intervention reveal

Intervention. Critical sample, then dextrose.

Glucose	pH	PaCO2	HCO3	SBE
100	7.38	38	22	-1
Na/Cl	Lactate	b-OHB		
138/105	1.5	0.2		

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-22 = 11) - 1.5 = 9.5 (normal).

Moderator take-home (final 1 min)

Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.

Case 13 - Shocked 2-week-old boy

Small-group phase (8 min) - you run this

Vignette. 2-week-old boy with 3 days of vomiting and poor feeding, now mottled and lethargic. Genitalia look normal. Weight below birth weight.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.27	28	12.5	-13	122
K	Cl	Glucose		
7.8	98	40		

Worked interpretation

Step	Answer
1. Clinical history	Neonatal shock with vomiting, poor feeding, mottling, lethargy, weight loss.
2. Internal validity	$[H^+] = 24 \times 28 / 12.5 = 54 \rightarrow$ pH 7.27 - matches.
3. pH	7.27 - acidaemia.
4. Primary disorder	HCO ₃ 12.5 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 12.5 + 8 = 26.75 \pm 2$. Actual 28 - appropriate.
6. Anion gap	$AG = 122 - (98 + 12.5) = 11.5$ (normal). NAGMA.
7. Delta ratio	Not applicable.
8. Oxygenation & integration	NAGMA with hyponatraemia, hyperkalaemia, hypoglycaemia $\rightarrow$ CAH salt-wasting crisis.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$122 - 98 = 24$	Markedly low - chloride-driven acidosis.
Atot	Albumin not given	-
Unmeasured anions	$11.5 - 0 = 11.5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Where groups go wrong

- Not recognising the triad of hyponatraemia, hyperkalaemia, and hypoglycaemia as CAH.
- Giving hydrocortisone before drawing the critical sample.
- Giving potassium-containing fluids to a hyperkalaemic neonate.

Your synthesis (final 2 min)

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. Draw the sample before steroids.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Classify the acid-base disorder and check compensation. Is this a HAGMA or a NAGMA?
- Q2. Three sidekick values make the diagnosis before any specialised test. Which, and what is it?
- Q3. A vomiting 2-week-old with pyloric stenosis would show which gas instead?
- Q4. Intervene: what are your first three actions, and which sample must precede them?

Answers

- A1. NAGMA with appropriate compensation.
- A2. Na 122 (hyponatraemia), K 7.8 (hyperkalaemia), glucose 40 (hypoglycaemia) $\rightarrow$ CAH.

A3. Metabolic alkalosis (hypochloraemic, hypokalaemic).

A4. (1) IV access, draw critical sample (electrolytes, glucose, cortisol, ACTH, 17-OHP, renin, aldosterone). (2) Give stress-dose hydrocortisone. (3) IV fluids with dextrose and sodium (normal saline bolus, then D5NS; no potassium initially due to hyperkalaemia). Sample must precede hydrocortisone.

Post-intervention reveal

Intervention. Hydrocortisone, fluids with dextrose, treat hyperkalaemia.

pH	PaCO2	HCO3	SBE	Na
7.35	35	18	-5	135
K	Cl	Glucose		
5.0	100	90		

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normalising). Unmeasured anions = AG (135-100-18 = 17) - 0 = 17 (raised - resolving).

Moderator take-home (final 1 min)

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.